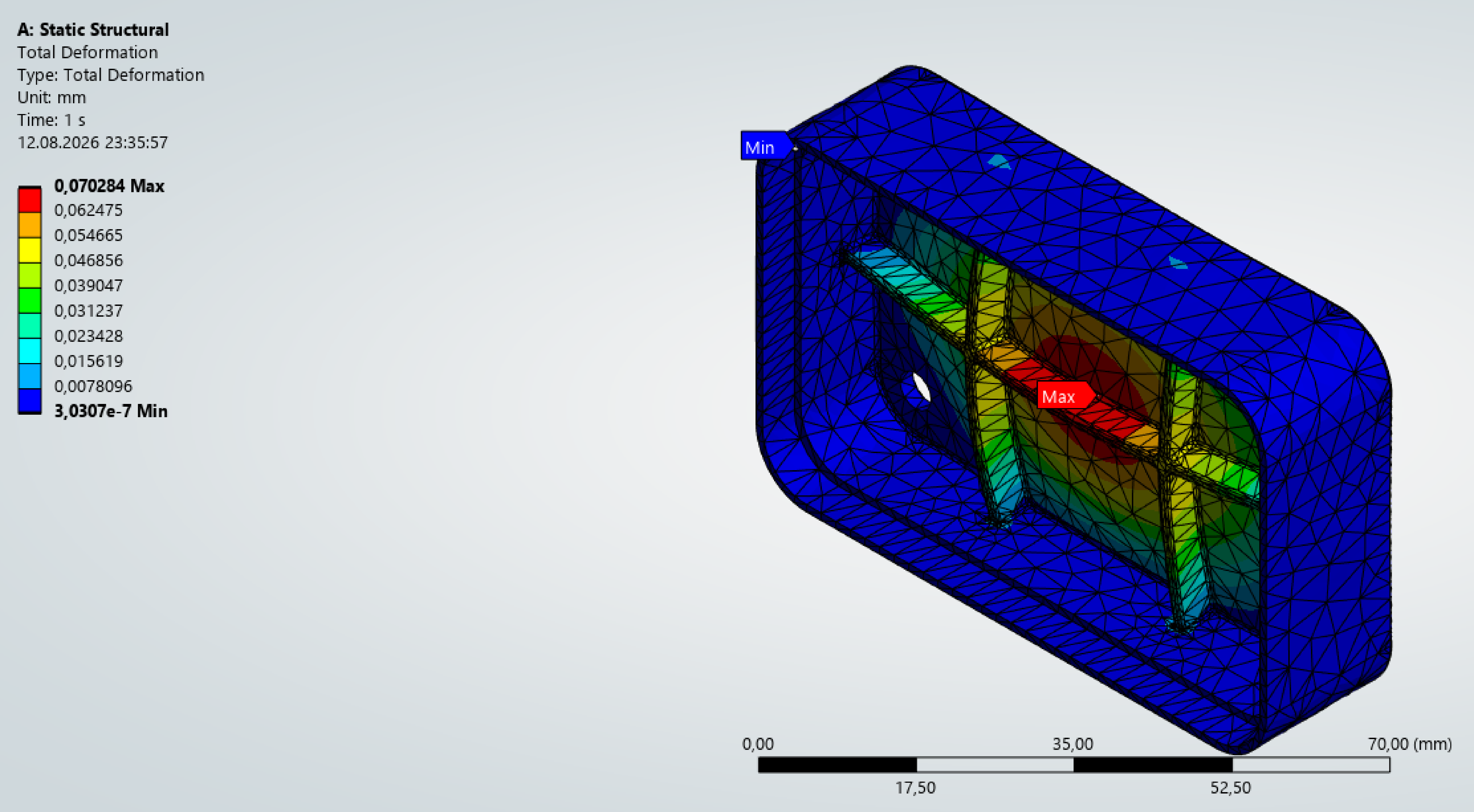
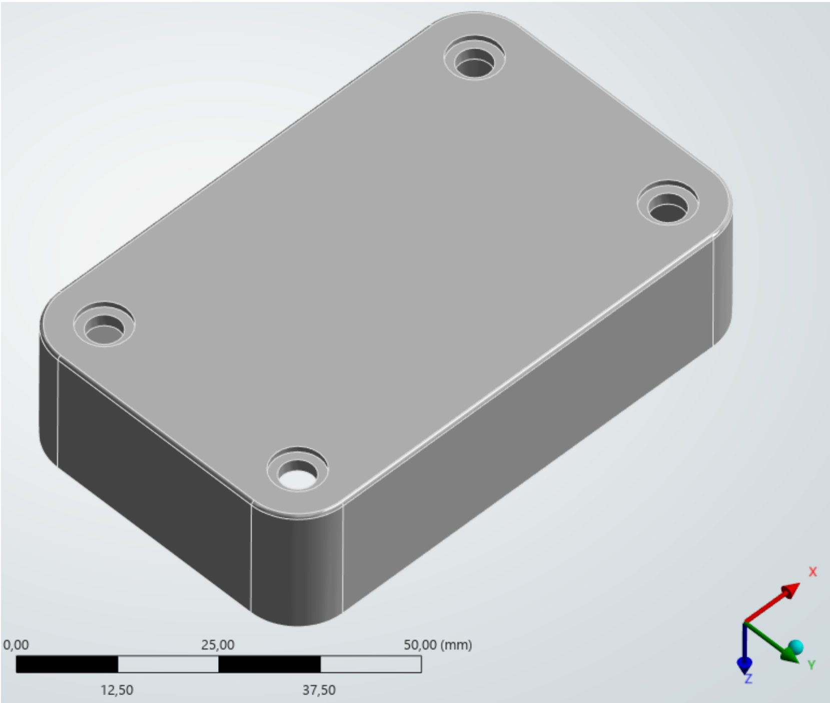
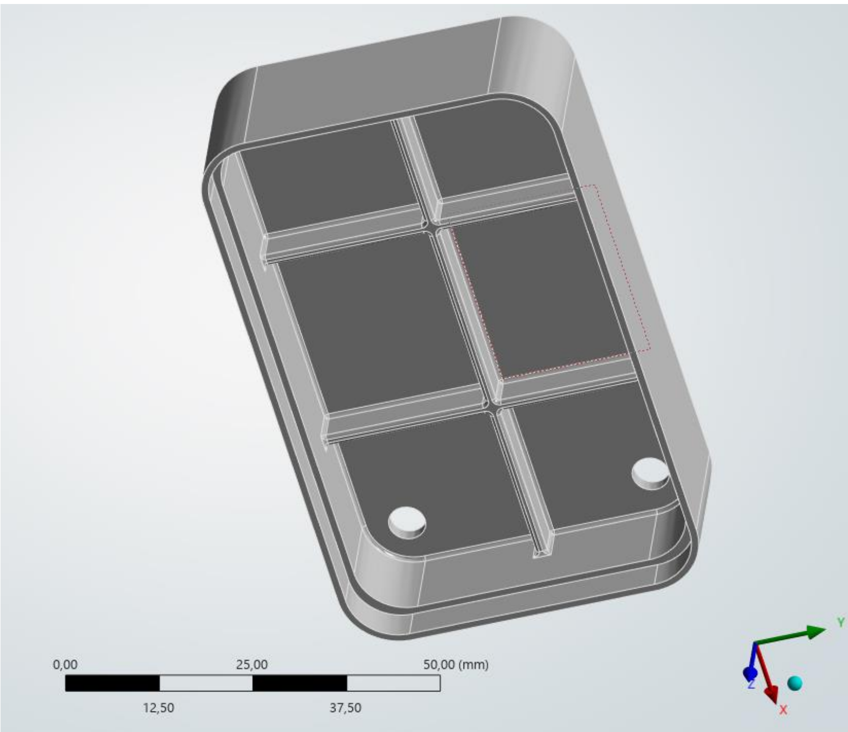

A: Static Structural
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1 s
12.08.2026 23:35:57
0,070284 Max
0,062475
0,054665
0,046856
0,039047
0,031237
0,023428
0,015619
0,0078096
3,0307e-7 Min
Min
Max
0,00
35,00
70,00 (mm)
17,50
52,50

## Static Structural Analysis & Mesh Sensitivity Study of an Aluminum Enclosure

## Project Objective

To evaluate the structural integrity, maximum equivalent stress, and safety factor of an aluminum enclosure cap under external pressure, and to provide engineering recommendations regarding its suitability for mass production.



## Boundary Conditions & Setup

The simulation was set up to replicate real-world assembly constraints and operational loads:

### Material

Aluminum Alloy Yield Strength: 280 MPa.

Outline of Schematic A2: Engineering Data

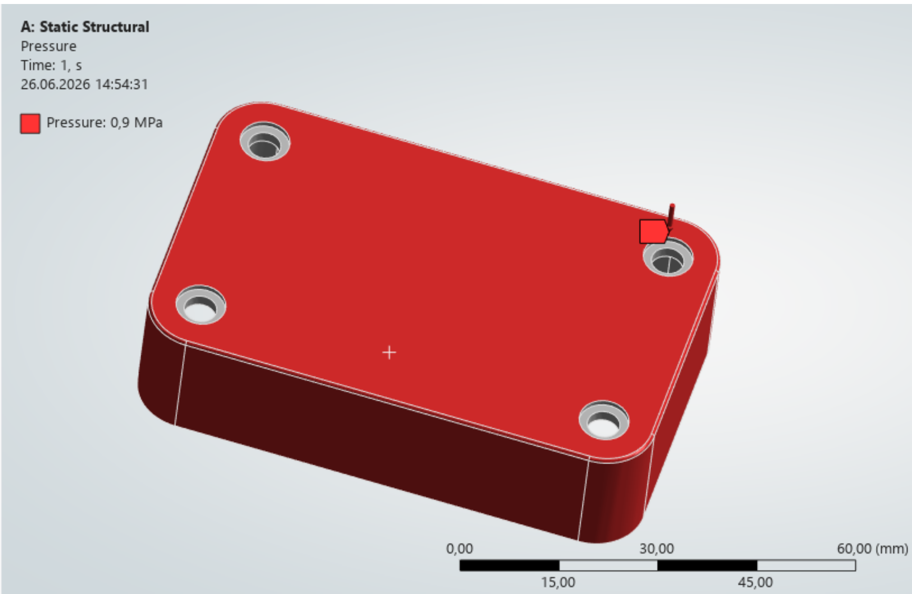
| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Contents of Engineering Data | | | Source | Description |
| 2 | Material | | | | |
| 3 | Aluminum Alloy | | | General_Materials.xml | General aluminum alloy. Fatigue properties come from MIL-HDBK-5H, page 3-277. |
| * | Click here to add a new material | | | | |

Properties of Outline Row 4: Aluminum Alloy

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Property | Value | Unit | | |
| 2 | Material Field Variables | Table | | | |
| 3 | Density | 2770 | kg m^-3 | | |
| 4 | Isotropic Secant Coefficient of Thermal Expansion | | | | |
| 5 | Coefficient of Thermal Expansion | 2,3E-05 | C^-1 | | |
| 6 | Isotropic Elasticity | | | | |
| 7 | Derive from | Young's Modulus and Poisson... | | | |
| 8 | Young's Modulus | 7,1E+10 | Pa | | |
| 9 | Poisson's Ratio | 0,33 | | | |
| 10 | Bulk Modulus | 6,9608E+10 | Pa | | |
| 11 | Shear Modulus | 2,6692E+10 | Pa | | |
| 12 | S-N Curve | Tabular | | | |
| 13 | Interpolation | Semi-Log | | | |
| 14 | Scale | 1 | | | |
| 15 | Offset | 0 | Pa | | |
| 16 | Tensile Yield Strength | 2,8E+08 | Pa | | |
| 17 | Compressive Yield Strength | 2,8E+08 | Pa | | |
| 18 | Tensile Ultimate Strength | 3,1E+08 | Pa | | |
| 19 | Compressive Ultimate Strength | 0 | Pa | | |

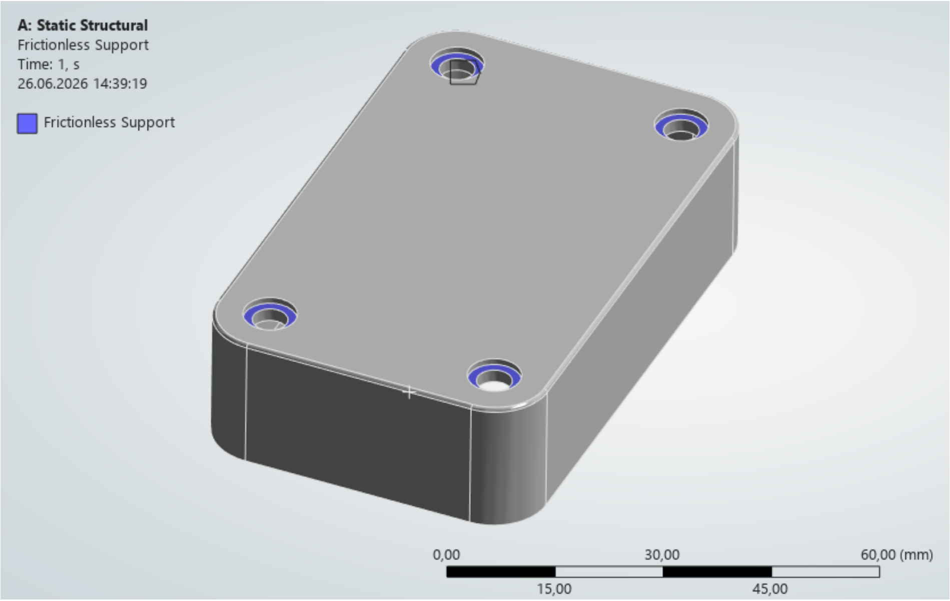
### Loading

A uniform external pressure of 0.9 MPa applied across 17 exterior faces.



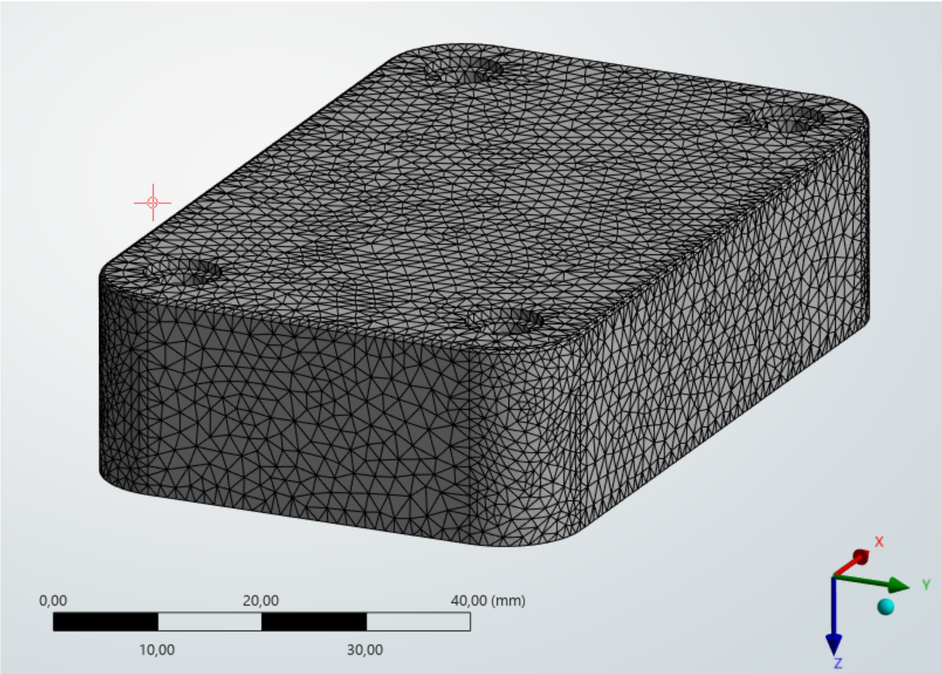
## Constraints

Frictionless supports applied to 4 mounting holes, the bottom contact surface, and internal walls. This conservative approach restricts normal displacement while allowing in-plane translation to simulate realistic assembly friction.



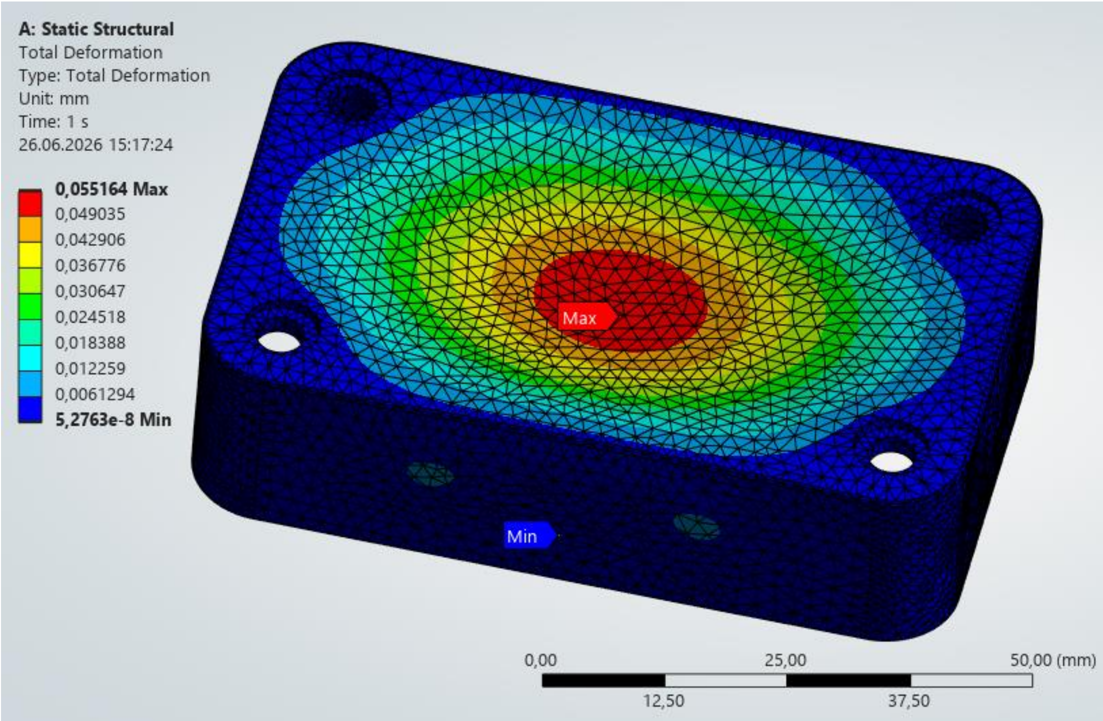

## Mesh Strategy

A mesh sensitivity study was conducted, comparing a default coarse mesh with a refined 0.95 mm element size to accurately capture local stress concentrations.

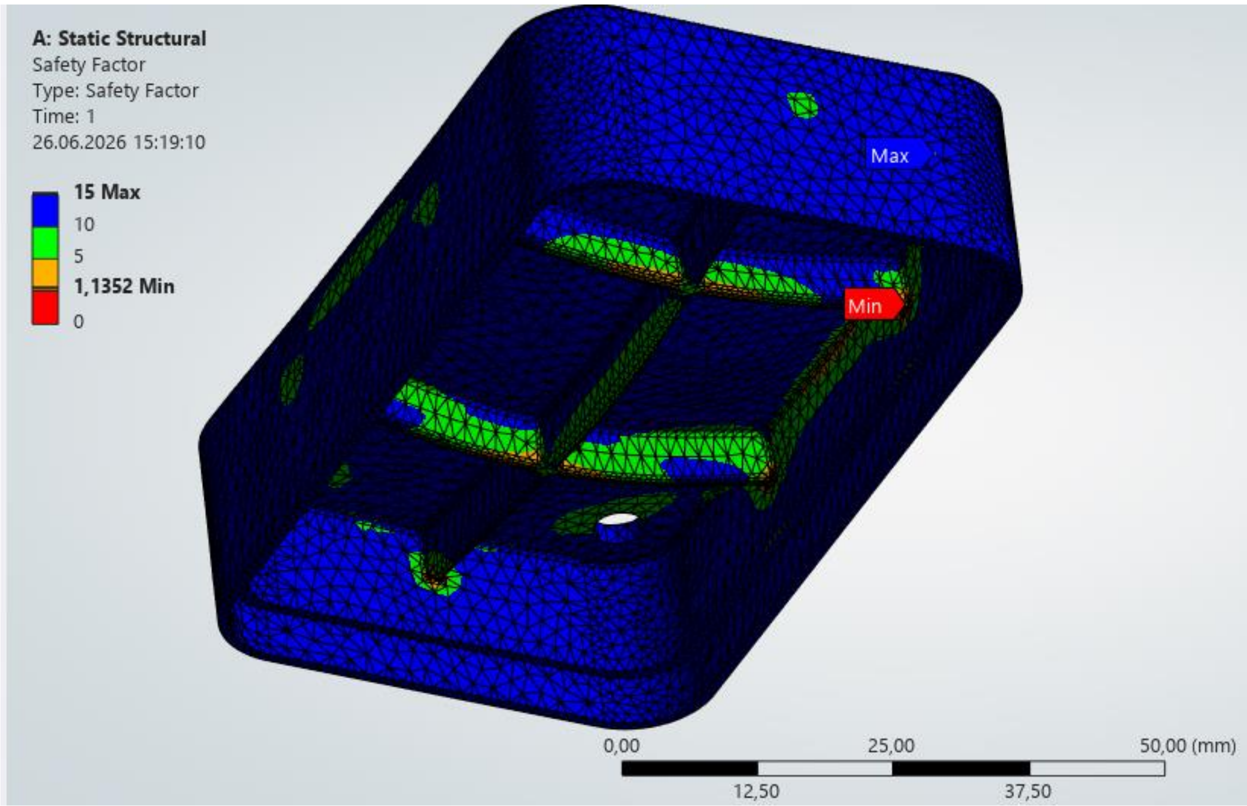


## Results & Engineering Conclusion

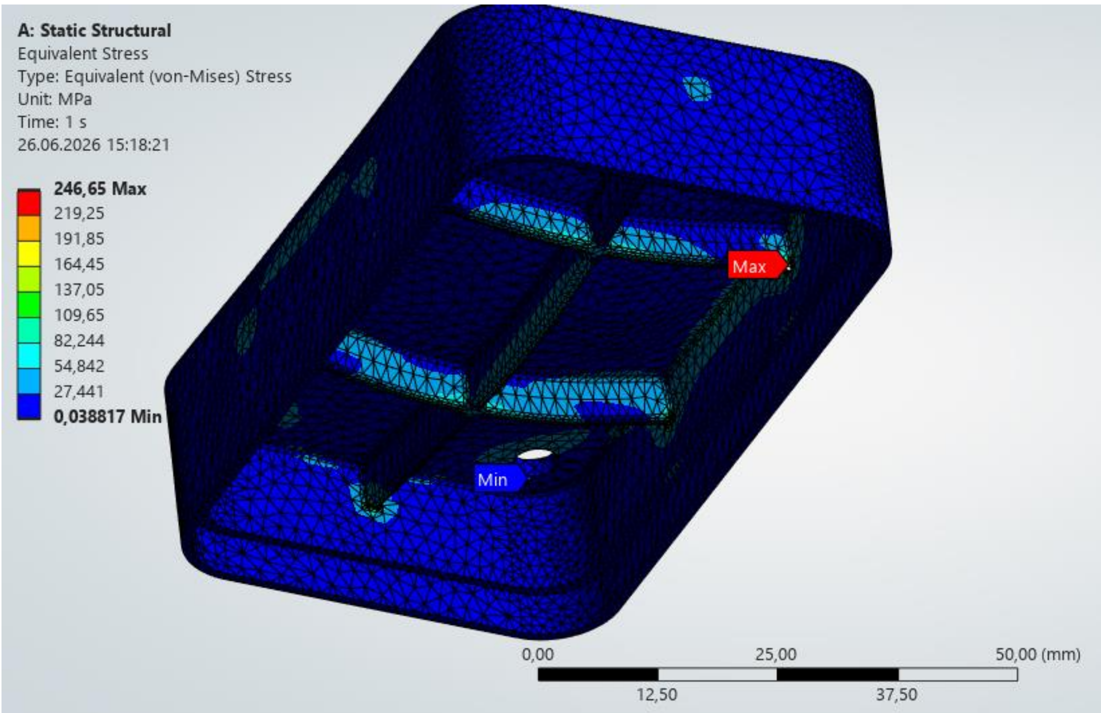
Maximum Total Deformation: 0.055 mm.




Minimum Safety Factor: 1.13.



Maximum Equivalent von Mises Stress: 246.5 Mpa.




## Summary

While the maximum calculated stress 246.48 MPa remains below the material's yield strength 280 MPa, keeping the part within the elastic deformation zone, the safety factor of 1.13 is critically low. Furthermore, the mesh refinement study revealed an 8.25% increase in peak stress compared to the coarse mesh, confirming severe local stress concentrations.

## Recommendation

It is strictly recommended to halt mass production approval for this current design. The geometry requires immediate optimization e.g., increasing wall thickness or expanding fillet radii to prevent irreversible plastic deformation under operational deviations or cyclic fatigue.

## Proposed Optimization

To achieve a target Safety Factor of >1.5 for safe mass production, the following design iterations were recommended:

Geometric Adjustments - Increasing the critical wall thickness and expanding the internal fillet radii to smooth out high-stress concentration zones.

Structural Reinforcement - Integrating internal support ribs stiffeners into the cap structure to better distribute the external pressure load without significantly increasing the part's total weight.